

CSCI351 Project 2

Java - Python - C# Programming Languages



PREPARED BY

Reham Issam Abou Taha

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Introduction

(1 page)

Java, Python, and C# are three influential programming languages, each offering distinct advantages and applications in the software development landscape. Java, known for its platform independence, allows developers to write code that runs on any device with a Java Virtual Machine. Python, celebrated for its simplicity and versatility, is widely adopted in web development, data science, and automation tasks. C#, supported by Microsoft's .NET framework, excels in developing Windows-based applications and games.

This document provides a detailed exploration of these three major programming languages: Java, Python, and C#. Each language offers unique features and strengths, making them suitable for various applications. By examining their definitions, workings, installation requirements, libraries, and characteristics, we aim to provide a comprehensive understanding of these languages.

1. Java

(1 page)

1.1. Definition:

Java is a high-level, class-based, object-oriented programming language designed to have as few implementation dependencies as possible. It is intended to let application developers write once and run anywhere, meaning that compiled Java code can run on all platforms that support Java without the need for recompilation.

1.2. How it works:

Java code is written in .java files. The code is compiled into bytecode using the Java Compiler. Finally, the bytecode runs on the Java Virtual Machine, which interprets it into machine code for the specific platform.

1.3. Libraries:

- Standard Libraries: java.util, java.io, java.net, java.sql, etc.
- Popular Third-Party Libraries: Apache Commons, Google Guava, Jackson for JSON processing, Hibernate for ORM, and Spring Framework for enterprise applications.

1.4. Characteristics:

- Platform-independent due to JVM.
- Strongly typed language.
- Robust standard library.
- Object-oriented.

1.5. Installation Requirements:

- JDK for writing and compiling Java code.

1.6. Development Environment:

- NetBeans: A popular IDE that provides a robust environment for developing Java applications.
- How to install: <https://youtu.be/45DcRMeLweE?si=yqp gn345aU8O0PMZ>

2. Python

(1 to 2 pages)

2.1.Definition:

Python is a high-level, interpreted, general-purpose programming language. Its design philosophy emphasizes code readability with the use of significant indentation.

2.2.How it works:

- Python code is written in .py files.
- The code is interpreted by the Python interpreter, which executes the commands directly.

2.3.Libraries:

- Standard Libraries: os, sys, math, datetime, json, etc.
- Popular Third-Party Libraries: NumPy for numerical computations, Pandas for data manipulation, Flask and Django for web development, TensorFlow and PyTorch for machine learning, Requests for HTTP requests.

2.4.Characteristics:

- Easy to learn and use due to simple syntax.
- Dynamically typed.
- Extensive standard library.
- Highly portable and extensible.
- Supports multiple programming paradigms (procedural, object-oriented, functional).

2.5.Installation Requirements:

- Python interpreter downloaded from python.org.
- IDE like PyCharm, VS Code, or even simple text editors like Sublime Text.

2.6.Development Environment:

- Anaconda: A distribution that simplifies package management and deployment, particularly useful for data science projects.
- How to install: https://youtu.be/itVBw_rNYjI?si=Ewrw8wCf-rUUSMvT
- Visual Studio Code:A versatile code editor that supports Python development with various extensions.
- How to install:https://youtu.be/9o4gDQvVklU?si=C_Dao8rRzQCKIvyZ

3. C# (C Sharp)

(1 to 2 pages)

3.1.Definition:

C# is a modern, object-oriented programming language developed by Microsoft as part of its .NET initiative. It is designed for developing a wide range of applications that run on the .NET Framework.

3.2.How it works:

C# code is written in .cs files.The code is compiled into an Intermediate Language by the C# compiler.Finally,The IL runs on the Common Language Runtime, part of the .NET Framework, which converts it into machine code.

3.3.Libraries:

- Standard Libraries: System, System.Collections, System.IO, System.Net, System.Linq, etc.
- Popular Third-Party Libraries: Newtonsoft.Json for JSON processing, Entity Framework for ORM, NLog and Serilog for logging, NUnit and xUnit for testing.

3.4.Characteristics:

- Part of the .NET ecosystem.
- Strongly typed and statically typed.
- Rich standard library and framework support.
- Object-oriented with support for modern programming concepts like asynchronous programming and Language Integrated Query.

3.5.Installation Requirements:

- .NET SDK for writing and compiling C# code.
- Visual Studio or Visual Studio Code for development.

3.6.Development Environment:

Visual Studio:

- Comprehensive IDE by Microsoft for C# and .NET development, ideal for large-scale projects with extensive debugging and project management tools.
- Favoured for complex C# projects due to its rich feature set.
- How to install: https://youtu.be/8sVKmwQ-ykg?si=961FBbU_Sdo8ZAut

Visual Studio Code (VS Code):

- VS Code is a lightweight, customizable code editor with strong support for C# development, offering features like syntax highlighting, IntelliSense, and debugging capabilities.
- While lacking some of Visual Studio's advanced features, it's suitable for various programming tasks and appeals to developers who prefer a minimalist environment or work on smaller-scale projects.
- How to install: <https://youtu.be/GVmtPO-UEps?si=O0MBnsrNjSMT8Lxm>

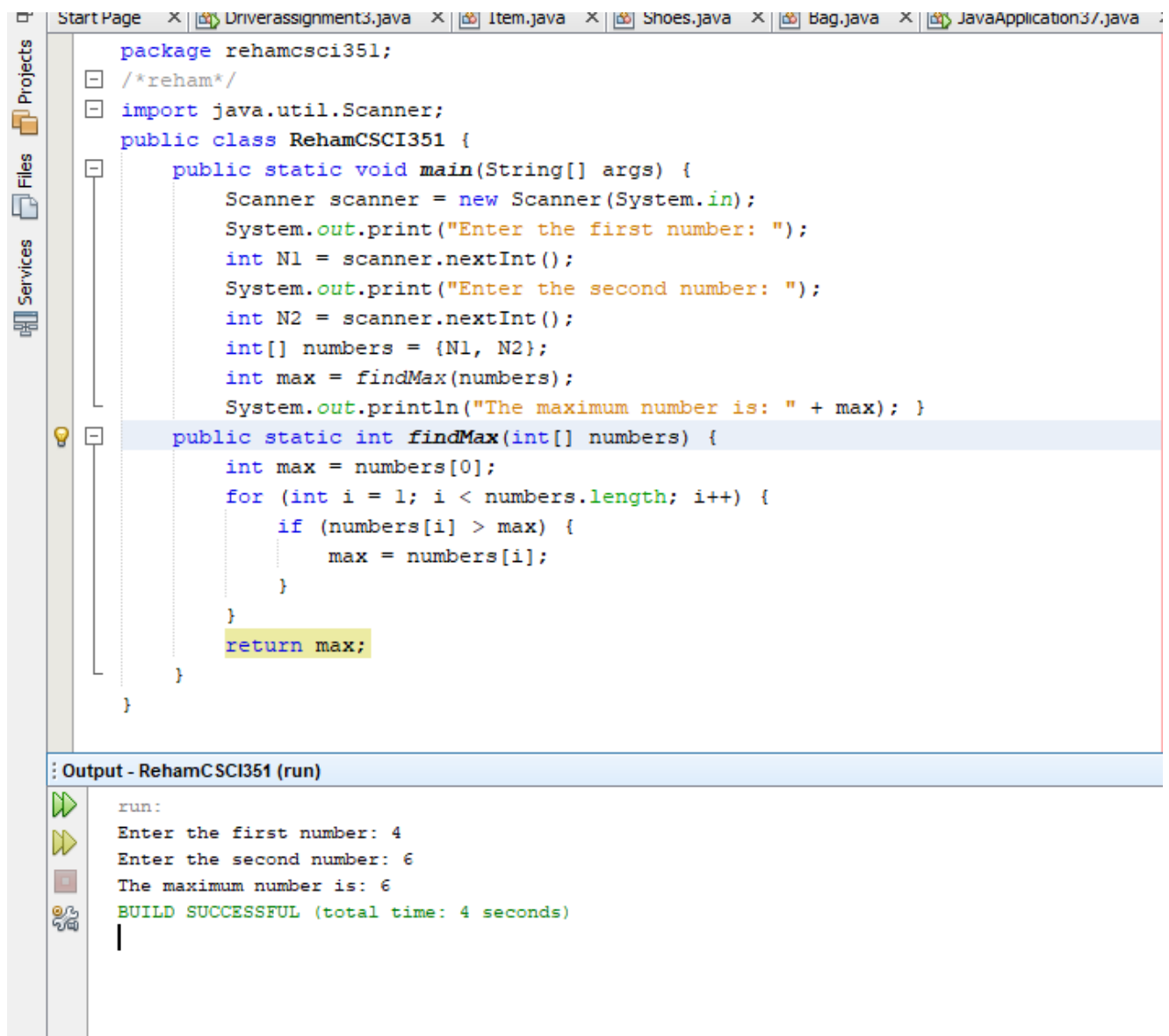
4. Some Online Compiler for these 3 languages:

- Replit
- Programiz
- Jdoodle
- OnlineGDB

5. Comparative Overview: Java, Python, and C# Programming Languages :

- High-Level Languages: Java, Python, and C# are high-level programming languages.
- Object-Oriented Principles: They support object-oriented programming (OOP) concepts.
- General-Purpose Usage: Suitable for various applications like web development, desktop software, and data processing.
- Community and Ecosystem: Strong developer communities and extensive libraries enhance their functionality.
- Garbage Collection: Automatic memory management reduces memory leak risks.
- Modern Programming Features: Support for features like asynchronous programming and exception handling improves productivity.
- AI Support: Python is favored for AI and machine learning tasks due to its specialized libraries like PyTorch. While Java and C# also have frameworks and libraries (like Weka for Java and ML.NET for C#) for AI development, Python is more commonly used in this domain.
- Open Source Status: Java and Python are fully open-source, while C#, as a language developed by Microsoft, is not fully open-source. However, many components of the .NET ecosystem, such as the C# compiler and runtime, are open-source.
- Case Sensitivity: While Java and C# are case-sensitive in all aspects of their language syntax, Python is primarily case-sensitive in identifiers such as variable names, function names, and class names..

Part 1:Java Code+Run (Using Netbeans):



The screenshot displays the NetBeans IDE interface. The top toolbar shows icons for Start Page, Projects, Files, and Services. The main editor window contains the following Java code:

```
package rehamcsci351;

/*reham*/
import java.util.Scanner;

public class RehamCSCI351 {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter the first number: ");
        int N1 = scanner.nextInt();
        System.out.print("Enter the second number: ");
        int N2 = scanner.nextInt();
        int[] numbers = {N1, N2};
        int max = findMax(numbers);
        System.out.println("The maximum number is: " + max); }

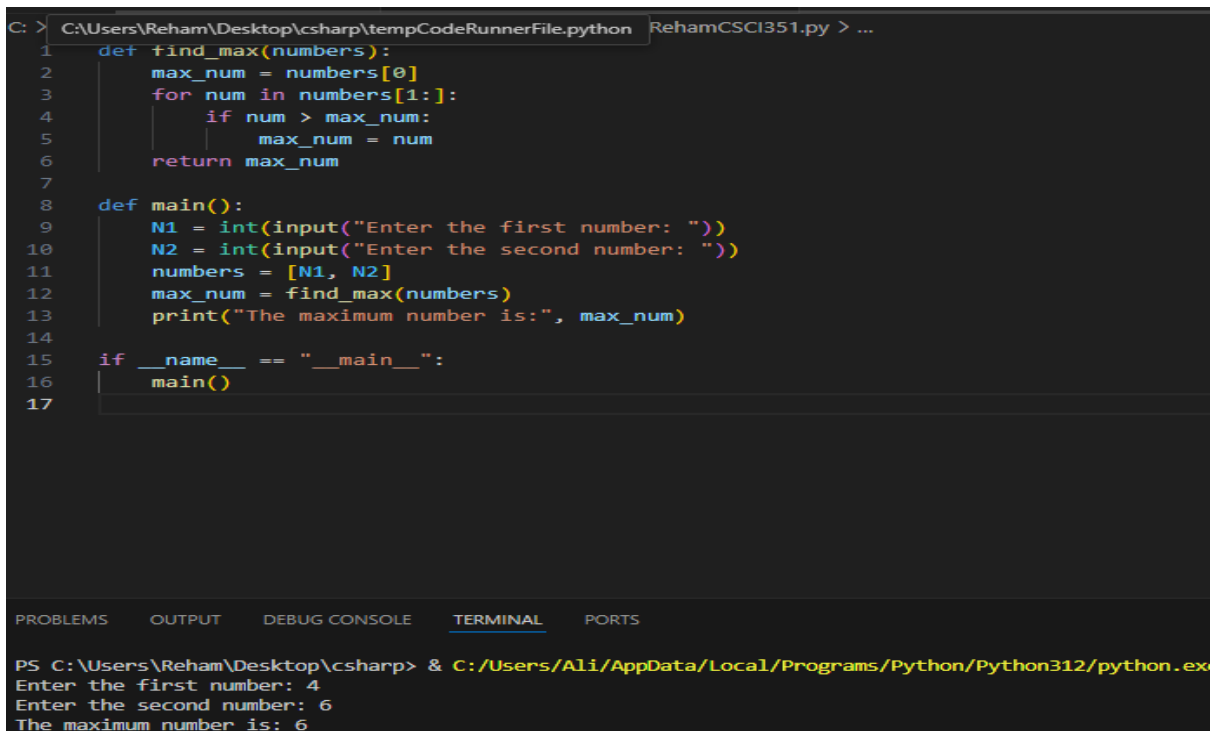
    public static int findMax(int[] numbers) {
        int max = numbers[0];
        for (int i = 1; i < numbers.length; i++) {
            if (numbers[i] > max) {
                max = numbers[i];
            }
        }
        return max;
    }
}
```

Below the code editor, the 'Output - RehamCSCI351 (run)' window shows the execution results:

```
run:
Enter the first number: 4
Enter the second number: 6
The maximum number is: 6
BUILD SUCCESSFUL (total time: 4 seconds)
```

Part 2: Question 1:

a. Python Code+Run (Using Visual Studio Code) :



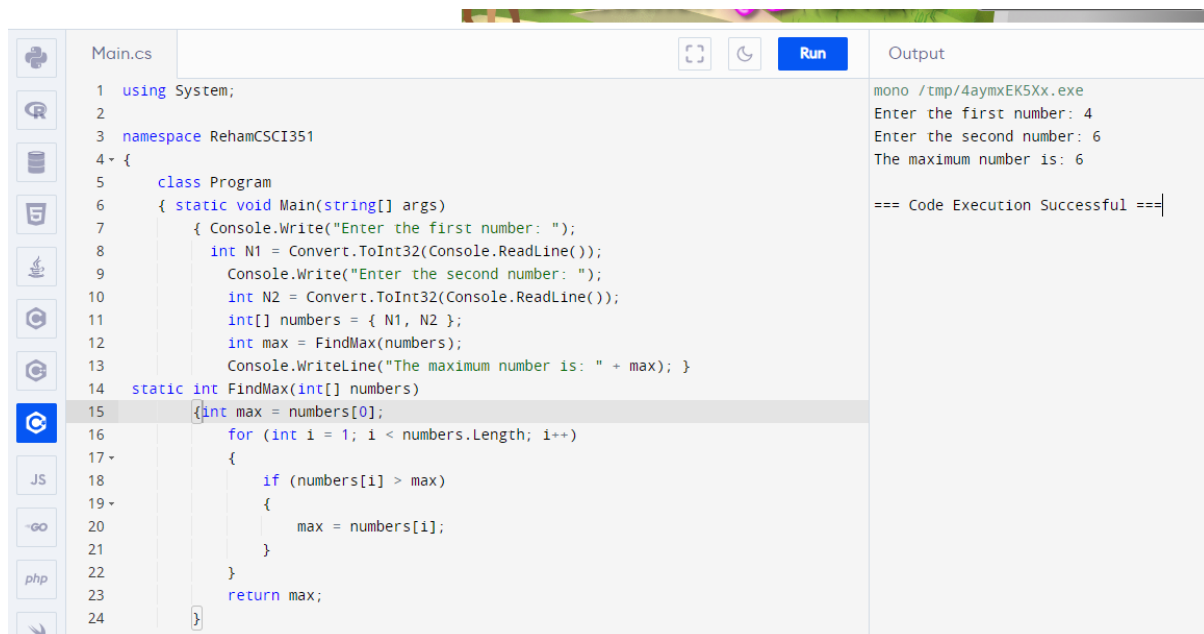
The screenshot shows the Visual Studio Code editor with a Python file named 'RehamCSCI351.py'. The code defines a function 'find_max' to find the maximum of two numbers and a 'main' function to take user input and call 'find_max'. The terminal at the bottom shows the command to run the script, followed by the input '4' and '6', and the output 'The maximum number is: 6'.

```
C:\Users\Reham\Desktop\csharp\tempCodeRunnerFile.python RehamCSCI351.py > ...
1 def find_max(numbers):
2     max_num = numbers[0]
3     for num in numbers[1:]:
4         if num > max_num:
5             max_num = num
6     return max_num
7
8 def main():
9     N1 = int(input("Enter the first number: "))
10    N2 = int(input("Enter the second number: "))
11    numbers = [N1, N2]
12    max_num = find_max(numbers)
13    print("The maximum number is:", max_num)
14
15 if __name__ == "__main__":
16     main()
17
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Reham\Desktop\csharp> & C:/Users/Alf/AppData/Local/Programs/Python/Python312/python.exe
Enter the first number: 4
Enter the second number: 6
The maximum number is: 6

b. C# Code+Run (Using Programiz):



The screenshot shows the Programiz online C# compiler. The code in 'Main.cs' defines a 'Program' class with a 'Main' method that takes two numbers as input and finds the maximum. The output window shows the execution results, including the input '4' and '6', and the output 'The maximum number is: 6'. A message at the bottom states '=== Code Execution Successful ==='.

```
1 using System;
2
3 namespace RehamCSCI351
4 {
5     class Program
6     { static void Main(string[] args)
7         { Console.WriteLine("Enter the first number: ");
8           int N1 = Convert.ToInt32(Console.ReadLine());
9           Console.WriteLine("Enter the second number: ");
10          int N2 = Convert.ToInt32(Console.ReadLine());
11          int[] numbers = { N1, N2 };
12          int max = FindMax(numbers);
13          Console.WriteLine("The maximum number is: " + max); }
14     static int FindMax(int[] numbers)
15     { int max = numbers[0];
16       for (int i = 1; i < numbers.Length; i++)
17       {
18           if (numbers[i] > max)
19           {
20               max = numbers[i];
21           }
22       }
23       return max;
24     }
25 }
```

mono /tmp/4aymxEK5Xx.exe
Enter the first number: 4
Enter the second number: 6
The maximum number is: 6
=== Code Execution Successful ===

Question 2:Mapping:

Java	Python	C#
package rehamcsci351;		namespace RehamCSCI351 {
import java.util.Scanner;		using system;
public class RehamCSCI351 {		class Program
public static void main(String[] args) {	def main():	static void Main(string[] args)
Scanner scanner = new Scanner(System.in);		
System.out.print("Enter the first number: ");	N1 = int(input("Enter the first number: "))	Console.Write("Enter the first number: ");
int N1 = scanner.nextInt();		int number1 = Convert.ToInt32(Console.Read Line());
System.out.print("Enter the second number: ");	N2 = int(input("Enter the second number: "))	Console.Write("Enter the second number: ");
int N2 = scanner.nextInt();		int N2 = Convert.ToInt32(Console.Read Line());
int[] numbers = {N1, N2};	numbers = [N1, N2]	int[] numbers = {N1, N2};
int max = findMax(number1, number2);	max_number = find_max(number1, number2)	int max = FindMax(numbers);

Conclusion:

In summary, Java, Python, and C# are three major programming languages, each with distinct strengths. Java's platform independence, Python's versatility, and C#'s integration with .NET make them valuable tools. Learning about their fundamentals, installation, and features provides a solid starting point for aspiring developers.